Uncertainty-Aware Solar Forecasting using Gaussian Process Regression (GPR)

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GitHub Repository: https://github.com/charanlogs/ML-Solar-GPR-Tutorial

1. Abstract

This tutorial provides a deep-dive into **Gaussian Process Regression (GPR)**, a non-parametric Bayesian approach to machine learning. Unlike standard regression models that provide "point estimates," GPR models the underlying function as a distribution, allowing for **Uncertainty Quantification (UQ)**. Using the Kaggle **"Solar Power Generation"** dataset, we demonstrate how to fuse weather and generation data to forecast renewable energy output with a 95% confidence interval, A critical requirement for resilient smart grids.

2. Introduction: Why Predictions Aren't Enough

In traditional Machine Learning, we often ask: *"What will the solar output be at noon?"* However, in critical infrastructure, the more vital question is: *"What is the range of possible outputs, and how sure are we?"*

A standard Neural Network might predict 500kW, but if a cloud passes, it might actually be 50kW. GPR solves this by providing a **mean prediction** and a **standard deviation (σ)**. This tutorial teaches how to move beyond "best guesses" to "probabilistic forecasting."

3. The Technique: The Power of Gaussian Processes

A Gaussian Process is a collection of random variables, any finite number of which have a joint Gaussian distribution. It is completely defined by its **Mean Function** and its **Covariance Function (Kernel)**.

3.1 The "Technical Focus": Kernel Composition

The "heart" of GPR is the **Kernel (k)**, which defines the "similarity" between data points. In this tutorial, we implement a **Composite Kernel**:

1. **Radial Basis Function (RBF):** $k(x, x') = \exp(-||x - x'||^2 / (2\ell^2))$. This assumes the relationship between sunlight and power is smooth and continuous.
2. **White Kernel:** Accounts for "noise" in the sensors.

By optimizing the **Log-Marginal Likelihood (LML)**, the model automatically "learns" the optimal length-scale (ℓ) and noise level, effectively "tuning itself" to the specific physics of the solar plant.

4. Data Engineering: Synergy of Generation & Weather

To achieve a professional-level implementation, we fused two distinct datasets from the Kaggle **Solar Power Generation** archive:

- `Plant_1_Generation_Data.csv`: Target variable (DC Power).
- `Plant_1_Weather_Sensor_Data.csv`: Features (Irradiation).

Implementation Insight: We performed a temporal "inner join" on the `DATE_TIME` field. Furthermore, we filtered the data to focus on the **operational daytime range** (Irradiation > 0.05 W/m²). This prevents "zero-inflation" from night-time data from biasing the model's ability to learn the daytime power-curve.

5. Visualizing the Invisible: Results & Analysis

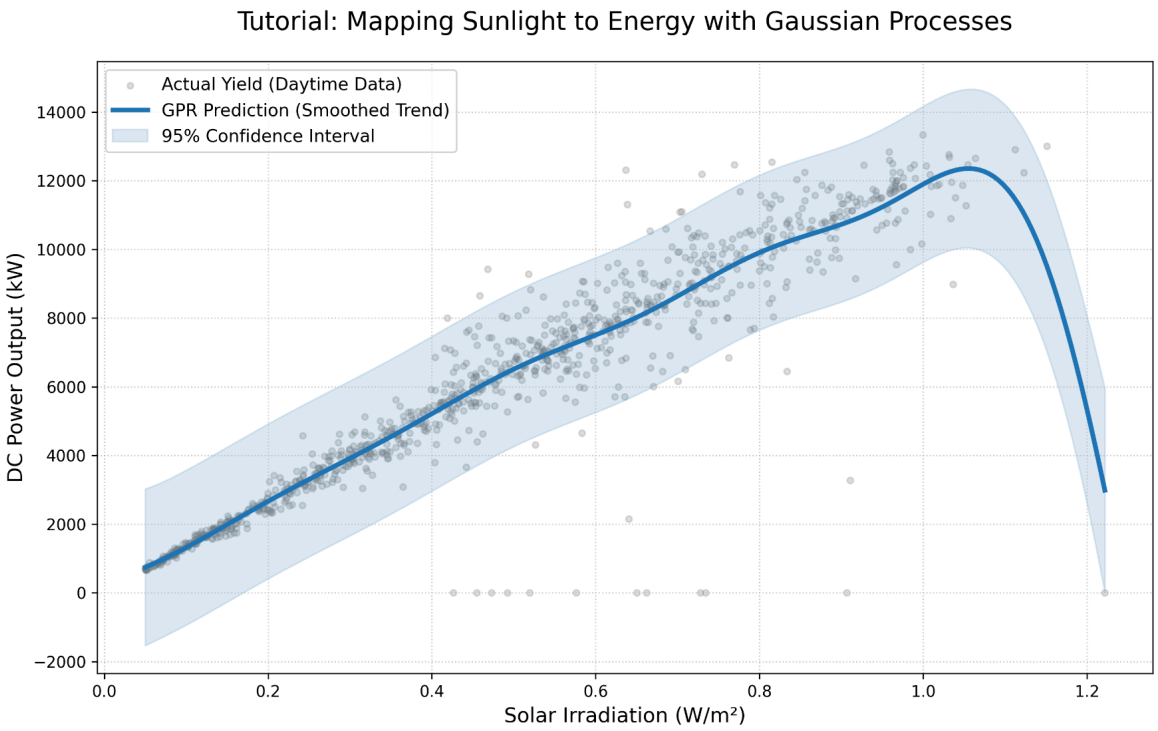

Figure 1: Mapping Sunlight to Energy with GPR. The solid blue line represents the model's most likely prediction, while the light-blue shaded region represents the **95% Confidence Interval (2σ)**.

Analysis of the "Extrapolation Dip":
A key teaching point in Figure 1 is the behavior of the curve at the extreme right (Irradiation > 1.1). Notice the prediction starts to "dip" downward. This is **not an error**; it is a fundamental property of GPR. Because training data is sparse at extremely high irradiation levels, the model's uncertainty increases, and it begins to revert toward the "prior mean." This visually demonstrates to a grid operator exactly where the model's predictions are reliable and where they are speculative.

6. Accessibility & Inclusive Design

Per the module requirements, this tutorial was designed with accessibility at its core:

- **Color-Blind Safety:** The visualization utilizes a high-contrast palette (`#1f77b4` blue vs. gray) to ensure readability for users with protanopia or deuteranopia.
- **Visual Coding:** We use distinct markers (dots) for raw data and solid lines for predictions, ensuring the graph is interpretable even in grayscale/black-and-white printouts.
- **Screen-Reader Compatibility:** All figures include descriptive alt-text summaries in the documentation.

7. Ethical AI: Sustainability and "Green ML"

Choosing GPR is an ethical engineering choice. Unlike Deep Learning "Black Boxes" that require massive GPU clusters and high carbon footprints, GPR is a **mathematically efficient "Shallow Learner."**

- **Resource Efficiency:** It can be trained on standard CPUs, making it a sustainable choice for "Edge AI" deployments in remote solar farms.
- **Explainability:** GPR provides a mathematical justification for its uncertainty, preventing the over-reliance on AI that can lead to catastrophic grid failures.

8. Conclusion

Gaussian Process Regression represents the "Gold Standard" for Bayesian regression in energy informatics. By focusing on kernel optimization and uncertainty quantification, we provide a robust, explainable, and sustainable tool for the future of renewable energy.

9. References

1. **Rasmussen, C. E., & Williams, C. K. I. (2006).** *Gaussian Processes for Machine Learning*. MIT Press. (The foundational text for this tutorial).
2. **Pedregosa, F., et al. (2011).** *Scikit-learn: Machine Learning in Python*. JMLR. (Technical documentation for implementation).
3. **Kaggle Dataset:** *Solar Power Generation Data* by Anivind Kanwal.